

Measuring the Drag Coefficient of a Sphere

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Research Question

When a sphere is dropped through air and reaches terminal velocity, does the experimentally measured drag coefficient match the theoretical value of $C_d = 0.47$ predicted for a sphere at Reynolds numbers between 10^3 and 10^5 ?

Introduction

When a sphere moves through air, it experiences a drag force described by the quadratic drag equation. The drag coefficient C_d characterizes the resistance of the object's shape - for a sphere, fluid dynamics predicts $C_d \approx 0.47$. This experiment aims to verify this prediction by dropping ping pong balls of different masses from a fixed height, measuring their speed, and calculating C_d experimentally.

Theory

When a sphere falls through air, it experiences a drag force opposing its motion:

$$F_{\text{drag}} = \frac{1}{2} \rho C_d A v^2$$

As the sphere accelerates, the drag force grows until it equals gravity. At this point the sphere reaches terminal velocity and stops accelerating:

$$mg = \frac{1}{2} \rho C_d A v^2$$

Solving for C_d : $C_d = \frac{2mg}{\rho A v^2}$ For a sphere, $C_d \approx 0.47$ is valid when the Reynolds number falls within $10^3 - 10^5$:

$$Re = \rho v \frac{D}{\mu}$$

For this experiment ($v \approx 9\text{--}12\text{ m/s}$, $D = 0.04\text{ m}$, $\mu = 1.85 \times 10^{-5}\text{ Pa}\cdot\text{s}$), $Re \approx 25,000 - 32,000$, confirming that $C_d \approx 0.47$ is the appropriate theoretical value.

Finite Drop Height Correction Because 12 m may not be sufficient for heavier balls to fully reach terminal velocity, a correction model was applied:

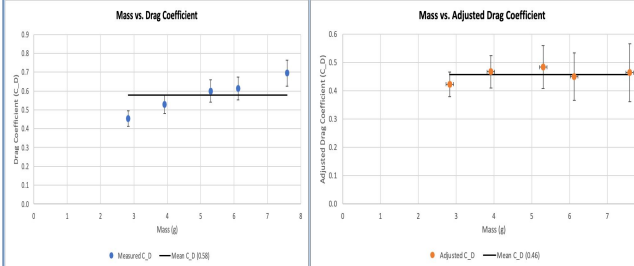
$$v_{\text{meas}}^2 = v_t^2 \left(1 - e^{-\frac{2gh}{v_t^2}}\right)$$

Apparatus

The experimental setup utilizes official 40 mm celluloid ping pong balls modified internally with varying quantities of sugar safely sealed with tape across the opening to prevent mass loss resulting in five distinct initial mass profiles (2.83g, 3.92g, 5.30g, 6.12g, 7.59 g) while maintaining a perfectly constant cross-sectional area (A). Mass measurements are precisely tracked before and after all trials using a mechanical triple beam balance to monitor potential shifting or loss of internal payload. Each sphere is released from rest from a height of 12 m inside a controlled indoor stairwell, where the descent data is captured using a tripod-stabilized smartphone recording video at 60fps positioned completely perpendicular to the vertical fall path. To convert pixel coordinates to physical displacement and eliminate optical parallax or lens distortion errors, a standard meter stick is placed perfectly within the vertical plane of the descent trajectory. Final frame-by-frame coordinate extraction and automated center-of-mass tracking are performed via *Tracker* open-source video analysis software.

Results

Figure 1 shows the directly measured C_d increasing systematically from 0.453 to 0.696 as ball mass increases, with a mean of 0.58. This trend indicates that heavier balls had not fully reached terminal velocity within the 12 m drop, as they require greater distance to do so.



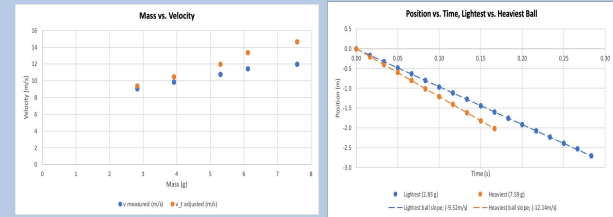
After applying the finite drop height correction, Figure 2 shows the adjusted C_d values clustering closely around a mean of 0.46, with no systematic trend with mass. The final experimental result is: $C_d = 0.458 \pm 0.078$

References

Taylor, J.R. *Classical Mechanics*. University Science Books, 2005. Section 2.4 — Quadratic Air Resistance.
Resnick, R., Halliday, D., Krane, K. *Physics*, 5th ed. Wiley, 2002.
Brown, D. *Tracker Video Analysis and Modeling Tool*. physlets.org/tracker. Accessed 2026.

Figure 3 confirms that the adjusted terminal velocity is consistently higher than the measured velocity, with the gap growing with mass.

Figure 4 shows the position vs. time for the lightest (slope = -9.52 m/s) and heaviest ball (slope = -12.14 m/s), both consistent with the correction model.



Discussion

Addressing Error;

We combined C_d scatter across the 5 masses with the propagated instrument uncertainty by root-sum-square, like the Pythagorean Theorem. The propagated error grew from 10 to 17 percent during the terminal velocity adjustment. In the 12m drop the balls don't reach terminal velocity (heaviest was 82% of v_t), and inverting the model amplifies any velocity error, and $C_d = 1/(v_t^2)$ doubles it again, hitting the heaviest ball the hardest (22% vs 10%). Ball to ball scatter was only 4%, well below the instrument uncertainty, because the dominant uncertainties (drop height, air densities), shift the ball rather than scattering.

Adjustment Explanation;

For adjusting the measured velocity to the terminal velocity, we used the equation

$$v_{\text{tn}} = v_t \sqrt{1 - e^{-2gh/v_t^2}}$$

Since it is impossible to isolate for terminal velocity, we calculated the measured velocity based on our drop height and terminal velocity values ranging from 6-18 m/s, with increments of 0.01. Then, for each ball, we matched the measured velocity to the terminal velocity producing it. This provided an accurate adjustment, however it increased uncertainty especially for the heavier balls.

Conclusions

The experimentally measured drag coefficient was found to be $C_d = 0.458 \pm 0.078$, compared to the theoretical value of 0.47, giving a percent error of 2.7%. This falls within the experimental uncertainty, confirming that the quadratic drag model accurately describes the motion of a sphere at the Reynolds numbers in this experiment ($Re \approx 25,000\text{--}32,000$). Without the finite drop height correction, C_d increased systematically from 0.453 to 0.696 with increasing mass. After applying the correction, all five values clustered around 0.46, demonstrating the importance of accounting for incomplete terminal velocity.

Main sources of uncertainty include video frame rate (60 fps), position measurement error, and air density variation.